

REAL-TIME IOT-BASED BEVERAGE BOTTLE LEVEL MONITORING SYSTEM

A PROJECT REPORT

Submitted by

MENAKA M
(23EIR062)

SIVACHAND S
(23EIR099)

THARUNKUMAR D
(23EIR111)

*in partial fulfilment of the requirements
for the award of the degree of*

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND INSTRUMENTATION

ENGINEERING

DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION

ENGINEERING



KONGU ENGINEERING COLLEGE

(Autonomous)

PERUNDURAI, ERODE - 638 060

OCTOBER 2026

**DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION
ENGINEERING**

KONGU ENGINEERING COLLEGE

(Autonomous)

PERUNDURAI, ERODE-638 060

OCTOBER 2026

BONAFIDE CERTIFICATE

This is to certify that the Project report entitled **REAL-TIME IOT-BASED BEVERAGE BOTTLE LEVEL MONITORING SYSTEM** is the Bonafide record of project work done by **MENAKA M (23EIR062)**, **SIVACHAND S (23EIR099)**, **THARUNKUMAR D (23EIR111)** in partial fulfilment of the requirements for the award of the Degree of Bachelor of Engineering in Electronics and Instrumentation Engineering of Anna University, Chennai during the year 2026 – 2027.

SUPERVISOR

HEAD OF THE DEPARTMENT

(Signature with seal)

Date:

Submitted for the end semester viva voce examination held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

**DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION
ENGINEERING**

KONGU ENGINEERING COLLEGE

(Autonomous)

PERUNDURAI, ERODE-638060

OCTOBER 2026

DECLARATION

We affirm that the Project report titled **REAL-TIME IOT-BASED BEVERAGE BOTTLE LEVEL MONITORING SYSTEM** being submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Date:

**MENAKA M
(23EIR062)**

**SIVACHAND S
(23EIR099)**

**THARUNKUMAR D
(23EIR111)**

I certify that the declaration made the above candidates is true to the best of my knowledge.

Date:

Name & Signature of the Supervisor with seal

ACKNOWLEDGEMENT

We express our sincere thanks and gratitude to our respected Correspondent **Thiru. E.R.K. KRISHNAN., M. Com**, and all other philanthropic trust members Kongu Vellalar Institute of Technology Trust who have always encouraged us in academic and co-curricular activities.

We wish to express our profuse thanks to our beloved and eminent Principal **Dr. R. PARAMESHWARAN., M.E., Ph.D.**, for his consistent support.

We take extreme pleasure in extending our warm respectful gratitude to our Head of the Department, **Dr. S.J. SUJI PRASAD M.E., Ph.D.**, for his valuable suggestions while working on this project.

We are immensely elated to proffer our sincere thanks to our esteemed guide **Dr. M. SASIREKA, M.E., Ph.D.**, for his valuable suggestions and guidance throughout the project duration.

We advance our gratitude to our **Project Coordinators and Review Committee Members**, Department of Electronics and Instrumentation Engineering for their valuable suggestions during the tenure of the project work.

We also thank **our faculty members, staff members, family members and our student friends** who have made this project possible.

ABSTRACT

In beverage manufacturing industries, maintaining the correct liquid level in every bottle is essential for ensuring product quality, reducing material wastage, and improving production efficiency. Traditional manual inspection methods are time-consuming, inconsistent, and prone to human errors, particularly in high-speed production environments. Overfilled and underfilled bottles may pass through the production line without being identified, which can lead to product rejection, increased production costs, and customer dissatisfaction. This project proposes a Real-Time Beverage Bottle Level Monitoring System using an ESP32-CAM and image processing techniques to automatically inspect the liquid level of bottles. The ESP32-CAM, equipped with an OV2640 camera, captures images of bottles moving on a conveyor system. An IR sensor is used to detect the arrival of each bottle and trigger image capture, while an LED provides uniform illumination for reliable image analysis. The captured images are processed using OpenCV, where image processing techniques are applied to identify the bottle and determine the liquid height. A calibration process is carried out using sample bottles with known fill levels to establish a suitable reference level and acceptable operating range. The system calculates the bottle fill percentage using the detected liquid height and bottle height. Based on the measured level, bottles are classified as underfilled, correctly filled, or overfilled. Bottles that do not meet the required filling range can be identified for segregation, while an alarm or indication can be generated for defective bottles. The proposed system provides a low-cost and efficient approach for real-time bottle inspection compared with manual inspection and expensive industrial vision systems. By continuously monitoring bottle levels, it helps minimize filling errors, reduce product wastage, improve inspection consistency, and increase overall production efficiency. The system can also be adapted to different bottle sizes and required filling levels through calibration, making it suitable for practical beverage production environments.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	v
	LIST OF TABLE	viii
	LIST OF FIGURES	ix
	LIST OF ABBREVIATIONS	x
1	INTRODUCTION	1
	1.1 INTRODUCTION	1
	1.2 EXISTING PROBLEM	2
	1.3 PROPOSED METHOD	2
	1.4 SUMMARY	3
2.	LITERATURE SURVEY	4
	2.1 INTRODUCTION	4
	2.2 LITERATURE SURVEY	4
	2.3 SUMMARY	7
3.	HARDWARE DESCRIPTION	8
	3.1 INTRODUCTION	8
	3.2 ESP32 CAM	9
	3.3 IR SENSOR	10
	3.4 CONVEYOR SETUP	11
	3.5 MOTOR DRIVER	12
	3.6 DC MOTOR	13
	3.7 SUMMARY	13
4.	METHODOLOGY	14
	4.1 INTRODUCTION	14

	4.2 CIRCUIT DIAGRAM FOR THE PROPOSED WORK	14
	4.3 BLOCK DIAGRAM FOR THE PROPOSED WORK	16
	4.4 FLOWCHART FOR THE PROPOSED WORK	17
	4.5 EXPERIMENTAL WORK	19
	4.6 SUMMARY	21
5.	RESULTS AND DISCUSSION	22
	5.1 INTRODUCTION	22
	5.2 RESULTS AND DISCUSSION	22
	5.3 SUMMARY	24
6.	CONCLUSION AND FUTURE SCOPE	25
	6.1 CONCLUSION	25
	6.2 FUTURE SCOPE	25
	APPENDIX-1	26
	REFERENCES	29
	MAPPING OF PROJECT GOALS TO SDG	31

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	ESP32	9
3.2	IR SENSOR	10
3.3	CONVEYOR SETUP	11
3.4	MOTOR DRIVER	12
3.5	DC MOTOR	13
4.1	CIRCUIT DIAGRAM OF THE PROPOSED WORK	15
4.2	BLOCK DIAGRAM OF THE PROPOSED WORK	17
4.3	FLOWCHART OF THE PROPOSED WORK	18
4.4	EXPERIMENTAL OF THE PROPOSED WORK	20

CHAPTER 1

INTRODUCTION

This chapter presents an overview of real-time beverage bottle level monitoring and its importance in beverage manufacturing industries. It discusses the challenges caused by variations in liquid filling levels and the limitations of manual inspection methods. Inconsistent filling can lead to product wastage, rejection, increased production costs, and customer dissatisfaction. A low-cost automated system using ESP32-CAM and image processing is proposed for real-time detection of underfilled, correctly filled, and overfilled bottles

1.1 INTRODUCTION

The beverage manufacturing process requires accurate and consistent filling of liquid into bottles to maintain product quality and production efficiency. The liquid level in every bottle must be maintained within a specified range to avoid underfilling and overfilling. Even small variations in the filling level can affect the quantity of the product, increase material wastage, and reduce customer satisfaction. In conventional beverage industries, bottle filling inspection is often carried out manually or using expensive industrial inspection systems. Manual inspection can be slow, inconsistent, and prone to human errors, particularly when bottles are moving continuously on high-speed production lines. Defective bottles may not be identified immediately and can continue through the production process. To overcome these limitations, an automated real-time bottle level monitoring system is proposed. In this project, an ESP32-CAM with an OV2640 camera is used to capture images of bottles moving on a conveyor. An IR sensor detects the arrival of each bottle and triggers the image capture process, while an LED provides uniform illumination for accurate image processing. The captured images are analyzed using OpenCV to determine the liquid height and calculate the corresponding fill percentage.

1.2 EXISTING PROBLEM

In existing beverage bottle inspection processes, identifying the exact liquid level of every bottle can be difficult when inspection is performed manually. Operators need to continuously observe bottles moving along the production line and identify whether each bottle is properly filled. This method becomes less reliable as the production speed increases.

One major problem is the occurrence of underfilled and overfilled bottles. Underfilled bottles contain less liquid than the required quantity, while overfilled bottles contain excess liquid. If these variations are not detected at the appropriate stage, defective bottles may proceed further in the production line. This can result in product rejection, material wastage, increased production costs, and customer dissatisfaction.

Another limitation of conventional inspection is the dependency on human observation. Manual inspection may result in inconsistent decisions because of operator fatigue, variation in observation, and the high speed of continuous production. Expensive automated industrial vision systems can overcome some of these problems, but their cost may be unsuitable for small-scale industries.

1.3 PROPOSED METHOD

The proposed method implements a real-time beverage bottle level monitoring system using an ESP32-CAM and OpenCV-based image processing. The system continuously monitors bottles moving on a conveyor and determines whether each bottle is underfilled, correctly filled, or overfilled. The ESP32-CAM with OV2640 camera is mounted above the conveyor to capture images of the bottles. An IR sensor is positioned to detect the arrival of a bottle and trigger the camera for image capture. An LED is used to provide uniform lighting conditions, which helps improve the reliability of image processing. Before operation, the system is calibrated using approximately 10–15 sample bottles having different known fill levels. A correctly filled bottle is used as the reference, and an acceptable fill-level range is defined. OpenCV image processing techniques are then used to identify the bottle and detect the liquid height. The measured value is compared with the predefined target range. Bottles within the acceptable range are identified as correctly filled, while bottles below or above the specified range are classified as underfilled or overfilled.

1.4 SUMMARY

This project focuses on improving beverage bottle inspection by addressing the limitations of manual liquid-level measurement. It implements an ESP32-CAM-based real-time monitoring system with an IR sensor, LED illumination, and OpenCV image processing to detect the liquid level of bottles. The system classifies bottles as underfilled, correctly filled, or overfilled based on the measured liquid height. The proposed system provides a low-cost and efficient approach for automated bottle inspection, reducing manual intervention and improving inspection consistency. By detecting filling variations in real time, the system can help reduce defective bottles, minimize material wastage, and improve the overall efficiency and reliability of the beverage production process.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

This chapter reviews existing research and techniques related to bottle fill-level inspection and automated industrial vision systems. It focuses on studies involving digital image processing, computer vision, object detection, image segmentation, and real-time inspection methods used for identifying liquid levels in bottles. The chapter also examines the use of OpenCV and camera-based systems for detecting underfilled and overfilled bottles. These studies highlight the importance of accurate image processing and automated inspection for improving product quality, reducing manual errors, and increasing efficiency in beverage manufacturing industries.

2.2 LITERATURE SURVEY

Gonzalez and Woods (2018) presented fundamental techniques in digital image processing for analyzing and interpreting images. The study covers important methods such as thresholding, segmentation, and edge detection. These techniques can be applied to identify objects and measure liquid levels in bottle images. The work provides a strong foundation for developing image-processing-based inspection systems [1].

Szeliski (2022) discussed various computer vision algorithms and applications for object detection and automated visual analysis. The study explains methods for feature extraction, image understanding, and object recognition. These techniques support the development of automated industrial inspection systems where bottles need to be detected and analyzed continuously. The research provides a basis for applying computer vision to real-time bottle inspection [2].

Bradski and Kaehler (2020) described the use of OpenCV for computer vision and image-processing applications. The work includes techniques such as image preprocessing, contour detection, and object analysis. These methods are useful for identifying bottle boundaries and determining the liquid level from captured images. The study supports the implementation of OpenCV-based real-time bottle level inspection [3].

Kumar and Sharma (2023) investigated vision-based bottle fill-level inspection using image-processing techniques. Their study demonstrated the automatic identification of underfilled and overfilled bottles. The research showed that image-based inspection can reduce errors associated with manual inspection and improve the consistency of quality checking. The findings directly support the development of an automated bottle level monitoring system for production environments [4].

2.3 SUMMARY

The literature shows that digital image processing and computer vision techniques can be effectively applied for automated bottle inspection. Studies have demonstrated the usefulness of thresholding, segmentation, contour detection, feature extraction, and object detection for identifying bottle characteristics and measuring liquid levels. OpenCV provides suitable image-processing methods for real-time analysis, while camera-based inspection can identify underfilled and overfilled bottles with improved consistency compared with manual inspection.

CHAPTER 3

HARDWARE DESCRIPTION

3.1 INTRODUCTION

This chapter provides a detailed overview of the hardware components used in the real-time beverage bottle level monitoring system. The hardware setup of the proposed system is designed to detect the liquid level inside beverage bottles automatically during the production process. The main components of the hardware include the ESP32-CAM with OV2640 camera, IR sensor, LED light, conveyor arrangement, and alarm or segregation mechanism. These components work together to capture bottle images, detect bottle arrival, provide proper illumination, and support real-time bottle level inspection.

The hardware arrangement forms the physical layer of the system, enabling real-time image capturing, sensing, processing, and defect identification. The ESP32-CAM is positioned above the conveyor to capture images of the bottles as they pass through the inspection area. The IR sensor detects the arrival of each bottle and provides a trigger for image capture. The LED light provides uniform illumination so that the bottle and liquid level can be clearly identified.

The ESP32-CAM acts as the main image capturing and control unit of the system. It receives the bottle detection signal from the IR sensor and captures the image of the bottle. The captured image is processed using image processing techniques to determine the liquid height. The measured liquid level is compared with the predefined acceptable level range. If the bottle is underfilled or overfilled, the system generates an alarm or activates the segregation mechanism. This automatic inspection reduces manual effort and helps maintain consistent bottle filling quality. The proposed hardware arrangement is based on the components specified in the project PPT.

3.2 ESP32 CAM

The ESP32-CAM is the main controller and image capturing unit used in this project for real-time beverage bottle level monitoring. It consists of an ESP32 microcontroller and an OV2640 camera module, which enables the system to capture images of bottles moving on the conveyor. The ESP32-CAM is mounted above the conveyor so that the bottle can be viewed clearly during the inspection process. The ESP32-CAM receives the bottle arrival signal from the IR sensor and captures the required image at the appropriate time. The captured image is used for image processing to identify the liquid level inside the bottle. The system uses image processing techniques to determine the detected liquid height and compare it with the desired bottle fill level. The ESP32-CAM provides a compact and low-cost solution for automated visual inspection. It also supports Wi-Fi communication, which can be used for transmitting monitoring information to an IoT dashboard. The camera-based inspection helps reduce manual checking and provides faster detection of underfilled and overfilled bottles. The ESP32-CAM is shown in Figure 3.1.



Fig 3.1 ESP32 CAM

3.3 IR SENSOR

The IR sensor is used in this project to detect the arrival of each beverage bottle at the inspection position. It works by detecting the interruption of infrared radiation when a bottle passes through the sensing area. In the proposed system, the IR sensor is placed near the conveyor at a suitable position. When a bottle reaches the inspection area, the sensor detects its presence and sends a signal to the ESP32-CAM. This signal acts as a trigger for capturing the bottle image. The use of an IR sensor helps synchronize bottle movement and image capture. It prevents unnecessary image capturing when no bottle is present and ensures that each bottle is inspected at the appropriate position. This improves the efficiency of the real-time monitoring system. The IR sensor is shown in Figure 3.2.



Fig 3.2 IR Sensor

3.4 CONVEYOR SETUP

The conveyor setup is used to transport the beverage bottles through the inspection area. It provides continuous movement of bottles and allows each bottle to be inspected at a fixed position. The ESP32-CAM is mounted above the conveyor so that it can capture the image of each bottle when it reaches the inspection point. The IR sensor detects the arrival of the bottle and triggers the image capturing process. The conveyor therefore provides proper movement and positioning of bottles for automatic inspection. The use of a conveyor-based arrangement makes the system suitable for production-line applications. Each bottle can be inspected without stopping the overall production process. The conveyor setup is shown in Figure 3.3.



Fig 3.3 Conveyor setup

3.5 MOTOR DRIVER (L298N)

The L298N motor driver is used in this project to control the operation and direction of the DC motor. It is a dual H-bridge motor driver module that allows a microcontroller such as the ESP32-CAM to control motors that require higher current than the microcontroller can provide directly. The L298N receives control signals from the microcontroller and supplies the required power to the motor.

In this system, the L298N motor driver is used to control the conveyor motor. It allows the motor to be started, stopped, and operated in the required direction. The motor speed can also be controlled by applying suitable PWM signals to the enable input of the driver. By using the L298N, the motor can be safely interfaced with the control system, improving the reliability and efficiency of the automated bottle inspection process. The L298N motor driver is shown in Figure 3.4.



Fig 3.4 Motor Driver

3.6 DC MOTOR

The DC motor is used in this project to drive the conveyor system and move the beverage bottles through the inspection area. It is an electrical motor that converts electrical energy into mechanical rotational motion. The motor receives power through the L298N motor driver, which controls its starting, stopping, direction, and speed according to the control signals from the ESP32-CAM. In the proposed system, the DC motor provides continuous movement of bottles along the conveyor. This enables each bottle to reach the image-capturing position for liquid-level inspection. The controlled movement of the conveyor ensures proper positioning and smooth transportation of bottles during the inspection process. The use of a DC motor helps reduce manual handling and improves the overall automation and efficiency of the system. The DC motor is shown in Figure 3.5.



Fig 3.5 DC Motor

3.7 SUMMARY

This chapter described the hardware components used in the real-time beverage bottle level monitoring system, including the ESP32-CAM with OV2640 camera, IR sensor, LED light, conveyor setup, alarm or segregation mechanism, and power supply. Each component performs an important function in the automatic bottle inspection process. The ESP32-CAM captures bottle images, the IR sensor detects bottle arrival, and the LED provides uniform illumination for accurate image processing. The system identifies underfilled, correctly filled, and overfilled bottles based on the measured liquid level.

CHAPTER 4

METHODOLOGY

4.1 INTRODUCTION

This chapter describes the methodology used to design and implement a real-time beverage bottle level monitoring system. The system integrates image capturing, image processing, sensing, and conveyor control components to ensure accurate and reliable inspection of beverage bottles. The ESP32-CAM captures the image of the bottle and provides the required image data for identifying the liquid level. The system compares the detected liquid level with the predefined target level and determines whether the bottle is underfilled, correctly filled, or overfilled. The conveyor system moves the bottles through the inspection area using a DC motor controlled by an L298N motor driver. When a bottle reaches the inspection position, the camera captures its image for analysis. Based on the detected liquid level, the system can identify defective bottles and provide an appropriate indication or segregation action. This automated inspection approach reduces manual intervention, minimizes inspection errors, and improves the efficiency and consistency of the bottle filling process.

4.2 CIRCUIT DIAGRAM FOR THE PROPOSED WORK

The circuit diagram of this project illustrates the interconnection of all the hardware components used to implement real-time beverage bottle level monitoring. The main controller of the system is the ESP32-CAM, which coordinates the camera operation, sensor input, and conveyor control. The system consists of an ESP32-CAM, IR sensor, L298N motor driver, DC motor, LED light, and power supply unit.

The IR sensor is used to detect the arrival of a beverage bottle at the inspection position. When the bottle is detected, the sensor provides a signal to the ESP32-CAM to initiate image capturing. The ESP32-CAM captures the image of the bottle using its camera and processes the required image information to determine the liquid level.

The L298N motor driver is connected between the ESP32-CAM and the DC motor. Since the controller cannot directly drive the DC motor, the motor driver provides the required current and allows proper control of the conveyor motor. The ESP32-CAM sends control signals to the L298N to operate the motor and move the bottles through the inspection area.

An LED light is provided near the inspection area to maintain uniform illumination while capturing bottle images. Proper lighting helps improve the quality and consistency of the captured images. All the components are powered using a suitable power supply, and proper grounding and wiring are maintained to ensure reliable operation of the system.

The integration of the camera, sensing, processing, and motor control components enables smooth communication between all parts of the system. This circuit arrangement provides an efficient and reliable method for automated beverage bottle level inspection.

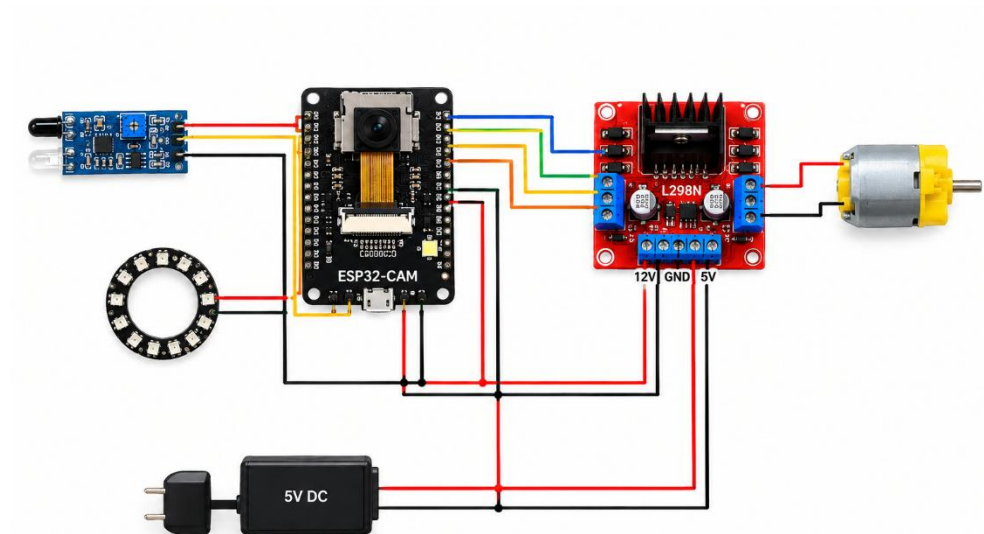


Fig 4.2 Circuit diagram for the proposed work

4.3 BLOCKDIAGRAM FORTHE PROPOSED WORK

The block diagram resents the overall architecture of the real-time beverage bottle level monitoring system. At the center of the system is the ESP32-CAM, which acts as the main image capturing and processing unit. A suitable power supply provides the required electrical energy to operate the ESP32-CAM and other connected components.

The conveyor system consists of a DC motor and L298N motor driver. The L298N receives control signals from the ESP32-CAM and controls the operation of the DC motor. The motor moves the beverage bottles along the conveyor towards the inspection area.

The IR sensor detects the presence of a bottle at the inspection position and sends a signal to the ESP32-CAM. Once the bottle is detected, the ESP32-CAM captures its image. The captured image is analyzed to determine the height of the liquid inside the bottle.

The measured liquid height is compared with the predefined target filling level. The system classifies the bottle as underfilled, correctly filled, or overfilled. If a defective bottle is detected, an alarm or segregation mechanism can be activated. The monitoring information can also be displayed through an IoT dashboard. Together, these components form a simple, reliable, and efficient system for automatic beverage bottle level monitoring and inspection. The block diagram of the proposed system is shown in Figure 4.2

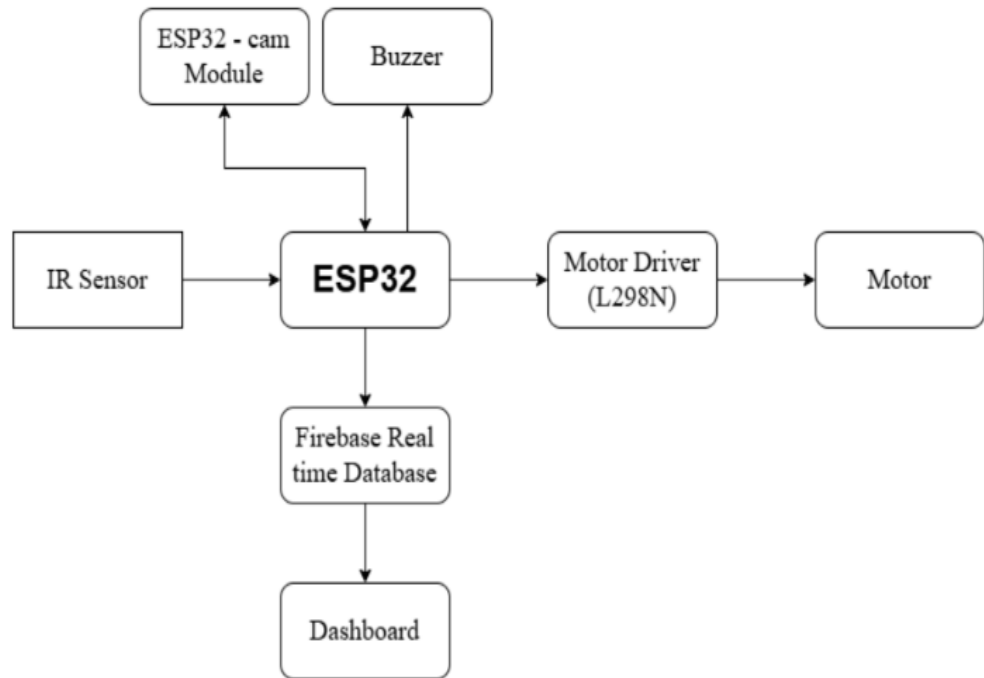


Fig 4.2 Block diagram for the proposed work

4.4 FLOWCHART FOR THE PROPOSED WORK

The block diagram represents the overall architecture of the real-time beverage bottle level monitoring system. At the center of the system is the ESP32-CAM, which acts as the main image capturing and processing unit. A suitable power supply provides the required electrical energy to operate the ESP32-CAM and other connected components.

The conveyor system consists of a DC motor and L298N motor driver. The L298N receives control signals from the ESP32-CAM and controls the operation of the DC motor. The motor moves the beverage bottles along the conveyor towards the inspection area.

The IR sensor detects the presence of a bottle at the inspection position and sends a signal to the ESP32-CAM. Once the bottle is detected, the ESP32-CAM captures its image. The captured image is analyzed to determine the height of the liquid inside the bottle.

The measured liquid height is compared with the predefined target filling level. The system classifies the bottle as underfilled, correctly filled, or overfilled. If a defective bottle is detected, an alarm or segregation mechanism can be activated. The monitoring information can also be displayed through an IoT dashboard. Together, these components form a simple, reliable, and efficient system for automatic beverage bottle level monitoring and inspection. The flowchart of the proposed system is shown in Figure 4.3.

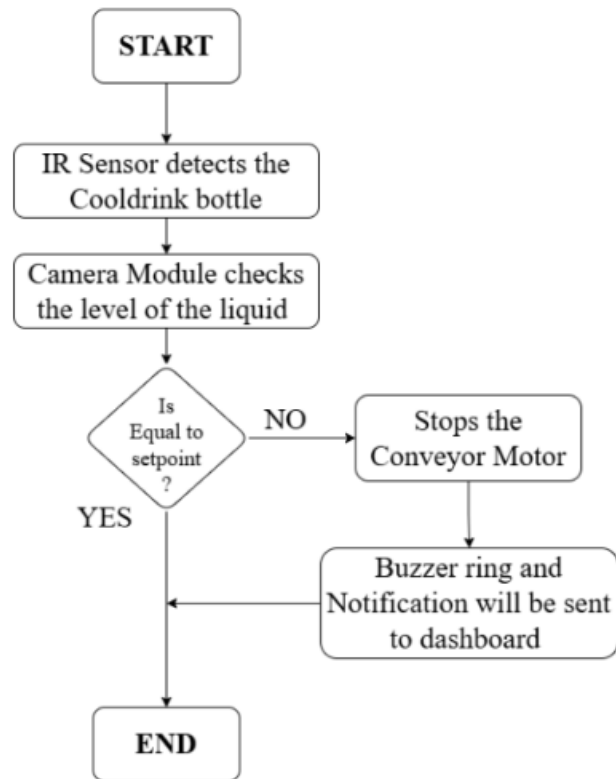


Fig 4.3 Flow chart for the proposed work

4.5 EXPERIMENTAL WORK

The experimental setup for the proposed project was developed to demonstrate real-time monitoring and inspection of beverage bottle filling levels using an ESP32-CAM-based system. The prototype consists of a conveyor arrangement used to transport beverage bottles through the inspection area. The conveyor is operated using a DC motor, which is controlled through an L298N motor driver.

The control hardware includes the ESP32-CAM, IR sensor, L298N motor driver, DC motor, and LED light. The ESP32-CAM receives the bottle detection signal from the IR sensor and captures the image of the bottle when it reaches the inspection position. The LED provides uniform illumination to obtain a clear image for level detection.

During the experiment, bottles with different liquid filling levels are placed on the conveyor. The bottles are moved towards the inspection area using the DC motor. When the IR sensor detects a bottle, the ESP32-CAM captures its image and analyzes the liquid level. The detected liquid height is compared with the predefined target level to classify the bottle.

Different bottle conditions such as underfilled, correctly filled, and overfilled bottles are tested to verify the operation of the system. The system identifies the filling condition and provides the corresponding output. Defective bottles can be indicated or separated from correctly filled bottles.

The experimental setup demonstrates that the proposed system can perform automatic bottle-level inspection with reduced manual intervention. The use of an ESP32-CAM, conveyor mechanism, IR sensor, and motor driver provides a low-cost and efficient approach for automated beverage bottle inspection. The experimental setup of the proposed system is shown in Figure 4.4.

Fig 4.4 Experimental setup for the proposed work

4.6 SUMMARY

This chapter presents the methodology and experimental setup of the real-time beverage bottle level monitoring system. The ESP32-CAM, IR sensor, L298N motor driver, DC motor, LED light, and conveyor system work together to perform automatic bottle inspection. The IR sensor detects the bottle, the ESP32-CAM captures and analyzes its image, and the conveyor system transports the bottles through the inspection area. The system determines whether the bottle is underfilled, correctly filled, or overfilled by comparing the detected liquid level with the predefined target level. The proposed methodology provides a simple, reliable, and efficient approach for automated beverage bottle level inspection while reducing manual effort and improving inspection consistency.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 INTRODUCTION

This chapter presents the results obtained from the experimental implementation of the real-time beverage bottle level monitoring system. The system performance is evaluated based on parameters such as bottle detection, liquid level measurement, classification of bottles, motor operation, and real-time monitoring. The ESP32-CAM captures the bottle image and processes the liquid level to identify whether the bottle is underfilled, correctly filled, or overfilled. The results obtained from the system are analyzed to validate the effectiveness of the proposed monitoring and segregation process.

5.2 RESULTS AND DISCUSSION

The experimental results demonstrate the successful implementation of the real-time beverage bottle level monitoring system. The ESP32-CAM continuously monitors the bottles moving on the conveyor and captures images when a bottle reaches the inspection area. The IR sensor detects the arrival of the bottle and triggers the image capturing process. The captured image is analyzed to determine the liquid level inside the bottle.

The system successfully identifies different bottle conditions based on the measured liquid height. The liquid level is compared with the predefined reference level to classify the bottle as underfilled, correctly filled, or overfilled. This automated inspection reduces the need for manual checking and helps maintain uniform filling quality during the production process.

The conveyor system is controlled using the L298N motor driver and DC motor. The motor moves the bottles continuously through the inspection area. When the IR sensor detects a bottle, the conveyor position allows the ESP32-CAM to capture the bottle image. The controlled movement ensures that bottles are inspected properly and reduces errors caused by improper positioning.

The measured value is compared with the target filling level. If the detected level is below the required range, the bottle is identified as underfilled. If it is within the specified range, it is identified as correctly filled. If the level exceeds the specified range, the bottle is identified as overfilled.

The system also provides an alert through the buzzer when an abnormal bottle condition is detected. This helps the operator identify defective bottles quickly. The automatic detection and alert mechanism reduce manual inspection time and improve the efficiency of the bottle inspection process.

The monitoring system also displays the inspection results through the Firebase real-time database and dashboard. The bottle status and corresponding inspection information can be viewed in real time. This allows the operator to monitor the production process and identify filling problems without continuously inspecting each bottle manually.

During the experimental testing, bottles with different liquid levels were passed through the conveyor. The system successfully detected the bottles and analyzed their filling conditions. The response of the ESP32-CAM, IR sensor, motor driver, DC motor, and buzzer showed that the proposed system can perform the required inspection process automatically.

Overall, the experimental results validate the performance of the proposed real-time beverage bottle level monitoring system. The system provides automatic bottle detection, liquid-level measurement, bottle classification, defective bottle identification, and real-time dashboard monitoring. The integration of image processing with the ESP32-CAM and automated conveyor operation provides a simple, low-cost, and effective solution for beverage bottle quality inspection, while reducing manual effort and improving production reliability.

5.3 SUMMARY

The results confirm that the proposed real-time beverage bottle level monitoring system successfully detects and monitors the liquid level of bottles. The ESP32-CAM captures bottle images and identifies underfilled, correctly filled, and overfilled bottles using image processing. The IR sensor detects bottle movement, while the L298N motor driver and DC motor provide controlled conveyor operation. The buzzer gives an alert for defective bottles, and the Firebase-based dashboard enables real-time monitoring of inspection results. Overall, the system reduces manual inspection, improves filling accuracy, and provides a simple and efficient solution for automated beverage bottle quality inspection.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The proposed Real-Time Beverage Bottle Level Monitoring System successfully monitors the liquid level of beverage bottles using the ESP32-CAM, IR sensor, L298N motor driver, DC motor, and image processing. The system identifies underfilled, correctly filled, and overfilled bottles automatically. The conveyor system provides smooth bottle movement, while the buzzer alerts the operator when an abnormal filling level is detected. The Firebase-based dashboard enables real-time monitoring of inspection results. Overall, the project demonstrates a simple, low-cost, and effective solution for automated bottle-level inspection, reducing manual intervention, minimizing inspection errors, and improving production efficiency.

6.2 FUTURE SCOPE

The future scope of this project includes extending the system to inspect additional bottle parameters such as cap position, bottle shape, label position, and other visible defects. Multiple cameras and improved image-processing techniques can be incorporated to increase inspection accuracy and support high-speed production.

The system can also be enhanced with advanced machine learning techniques for more accurate bottle classification under different lighting and operating conditions. An automatic rejection mechanism can be added to remove defective bottles from the conveyor without manual intervention.

Furthermore, the system can be integrated with PLC and HMI systems for industrial-scale applications. Cloud-based data logging and production analysis can also be improved to maintain inspection records and identify recurring filling problems. By upgrading the hardware and scaling the conveyor system, the proposed prototype can be developed into a reliable industrial automated bottle inspection system for beverage manufacturing.

APPENDIX-1

REFERENCES

1. Yazdi, L., Prabuwno, A. S., & Golkar, E. (2011). "Feature Extraction Algorithm for Fill Level and Cap Inspection in Bottling Machine." *2011 International Conference on Pattern Analysis and Intelligent Robotics (ICPAIR)*.
2. Shah, C., & Gund, N. (2020). "Bottling Line Inspection System Using Digital Image Processing." *International Conference on Advances in Science, Engineering and Technology (ICAST)*.
3. Sharma, S., Krupa, K. V., Gandhi, R., Jain, A., & Shah, N. (2015). "Empty and Filled Bottle Inspection System." *International Journal of Image, Graphics and Signal Processing / ICTACT Journal of Image and Video Processing*.
4. Pithadiya, K. J., Modi, C., & Chauhan, J. D. (2011). "Selecting the Most Favourable Edge Detection Technique for Liquid Level Inspection in Bottles." *International Journal of Computer Science and Information Management Applications*.
5. Pithadiya, K. J., Modi, C., & Chauhan, J. D. (2009). "Comparison of Optimal Edge Detection Algorithms for Liquid Level Inspection in Bottles." *2009 International Conference on Emerging Trends in Engineering and Technology (ICETET)*.
6. Yazdi, L. (2012). "Cap and Fill Level Inspection for Plastic Bottle Using Vision System." *Master's Thesis, Universiti Kebangsaan Malaysia*.

MAPPING TO SDG GOALS

The Project **“REAL TIME IOT BASED BEVERAGE BOTTLE LEVEL MONITORING SYSTEM”** aligns with the United Nations **Sustainable Development Goal (SDG) 9 - Industry, Innovation, and Infrastructure**: The implementation of smart instrumentation in the fly ash brick manufacturing process enables Industry 4.0 adoption by automating moisture monitoring and control. This enhances industrial efficiency, promotes innovation in manufacturing processes, and supports the development of intelligent infrastructure.

Sustainable Development Goal (SDG) 12 – Responsible Consumption and Production: The system reduces material waste by precisely controlling moisture levels and mixture proportions, minimizing defective bricks and rejections. This contributes to sustainable production practices and efficient resource utilization in brick manufacturing.